

NavigAR: Enhancing Localized Space Navigation using Augmented Reality

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ABSTRACT

In this study, we use Augmented Reality (AR) to project and let users interact with a 3D-replicated model of a fenced space to understand how AR can improve navigation and enhance space retention by improving wayfinding via recall. We recreated a fenced space in 3D using Blender. Users can see their current position and, using on-screen buttons, see routes to their destinations. We found that users could associate the 3D models of the buildings with their real-world counterparts. We observed that users were better able to navigate the campus after using the application.

Index Terms: Human-centered computing—Augmented Reality—Wayfinding instructions

1 INTRODUCTION

Augmented Reality (AR) has emerged as a transformative technology reshaping how we interact with technology and the real world. We aim to use a usability-focused approach towards building the application to ease users without prior exposure to AR. Users that use some sort of visual aid are less likely to make mistakes and take lesser time to navigate [4]. However, traditional turn-by-turn navigation systems hamper how humans spontaneously acquire store, retrieve, and communicate spatial information [2]. This means that even if the user is able to get to their destination using traditional systems, they are not able to do so independently without the aid of external navigational tools. Spatial chunking can be used to improve recall rates using landmarks that can help users remember and correlate their environment to their route better [2]. For a fenced space with few buildings such as a college campus, individual buildings with enough distinction can be considered as landmarks. By rebuilding these landmarks in 3D thus making a more representative referent would lead to better user performance for wayfinding [5].

2 LITERATURE REVIEW

We aim to build a digital map interface where users can interact in various levels of detail and avail directions to their desired destination with location-based context leveraging landmarks and guided navigation. One of the key problems discussed by all the papers we examined was that users would suffer from an information overload and face difficulty identifying spatial knowledge from Points of Interest (POIs) [1] [5] [6]. The various pain points noted from a User Journey Mapping study were summarised as follows [1]:

1. The POIs often did not match their 3D counterparts, leading to confusion.
2. Users faced overcrowding in their screens with too many POIs overlapping each other.

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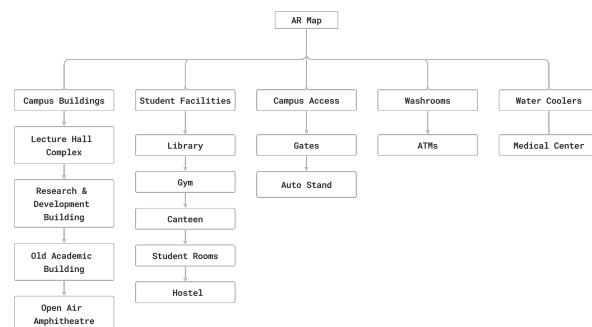


Figure 1: We decided to separate four features: Toilets, ATMs, Water Coolers and Medical Centre and let users have quick and easy access to them as we viewed these as the most relevant and important features.

3. The map lacked directional context, confusing the users about the direction in which the POIs would be located.

Various solutions to these problems were examined such as Virtual Global Landmarks (VGLs) [5] and better referents [1]. To ensure that our solution effectively tackles these challenges, we created a script wherein users underwent two rounds of testing, one with the 3D AR map and one post experiencing the 3D AR Map. This way, we could record the map's usability and look at the bigger goal, enhancing memorability based on previous studies [3].

3 METHODOLOGY

The principal aim of NavigAR is to alleviate the complexities and challenges associated with first-time navigation within the college campus. Leveraging Unity Software, we followed the below methodology:

1. Placing AR markers across pivotal points within the campus.
2. Developing a robust AR app that efficiently recognises these markers and displays a detailed 3D model of the campus and a marker to pinpoint the user's current location.
3. Crafting a user-friendly interface for users to select various locations and view routes to guide them to their destination.

3.1 Information Architecture

To avoid an information overload, we first created a rigid Information Architecture to distribute the features and let the user navigate the UI quicker as seen in Figure 1.

3.2 3D Development

We created 3D models of the campus from scratch using the design tool Spline. However, according to the free plan, we could only export .gltf files without colour. We then had to convert these .gltf files to .obj format and import them into Blender. We then imported

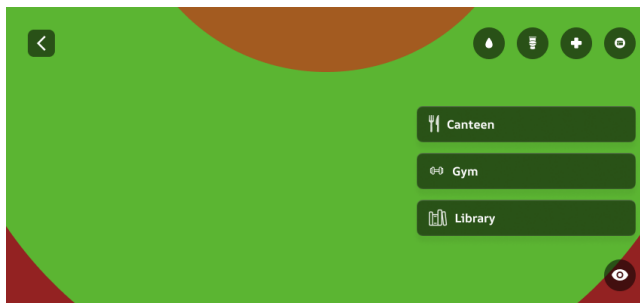


Figure 2: The red zones highlight areas harder to reach than the green zones where we kept our buttons.

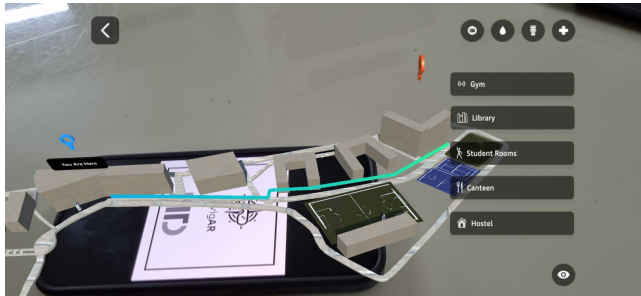


Figure 3: The UI of the application with a route highlighted.

this .obj file to Unity and created the working prototype. This model provides a comprehensive visual representation of the user's immediate surroundings. The 3D model enhances user experience and aids in spatial orientation, allowing users to understand where they are scanning from relative to other buildings and landmarks. Once the user scans the model, they are shown a pointer indicating their current position.

3.3 Interactivity and User Interface (UI) Development

The menu-driven program powered by C# scripts is at the heart of our navigation system. These scripts are responsible for the functionality of each button on the UI. When a user selects a destination, be it a specific building, medical centre, water cooler, or ATM, the corresponding C# script activates, guiding the user to their selected location. Based on the user's selection, the application highlights a path to the destination in a blue-green gradient reminiscent of our college colours.

Besides cleaning the clutter with an Information Architecture, we also wished to create and use an AR-friendly UI. To allow the user to view and learn the model better, we also give the option to hide the UI and focus solely on the model and the selected route. Typography is a crucial aspect of our UI design. Its low contrast, generous spacing, and robust shapes work well in busy backgrounds with high readability. The design of letterforms is based on research and thorough testing on various devices ranging from high-end headsets to low-resolution smartphone-based devices. Understanding that many users will operate NavigAR in landscape mode, we've designed the UI layout for optimal reachability. Buttons and interactive elements are thoughtfully positioned, enabling users to easily access them with one hand, making the navigation experience seamless.

4 RESULTS AND CONCLUSION

We tested the application with five students, three of whom were freshmen and an autorickshaw driver who had never seen the campus before. We found that users believe the model's level of detail is adequate to provide context to the buildings. Previously, we had kept the buttons on the left. However, while testing the app, we realised that the current button placement forces the user to obstruct the camera. Learning from this, we then shifted the buttons to the right. Students faced minor difficulty finding the buttons for their desired destinations on their first use; however, on their second use, they could easily navigate the UI. The autorickshaw driver struggled to navigate the UI in the first and second tries, showing that the UI is difficult for users with no campus context.

We conducted two rounds of testing; one while using the AR Map as a means of wayfinding and a second round of Independent wayfinding. All of the users, including the autorickshaw driver, were successfully able to reach their destination in both rounds of testing. All users showed a mostly positive attitude towards the concept. Most of the feedback we received pertained to the quality of the 3D models being on the lower side. Since this was a prototype, the buildings lacked clear visual distinction regarding colours and textures.

In conclusion, the NavigAR project successfully demonstrates the potential of Augmented Reality (AR) in enhancing campus navigation at IIIT Delhi. By leveraging AR technology, the project offers an intuitive, interactive, and user-friendly navigation experience, particularly for newcomers and visitors. Implementing AR markers and a model of the campus, combined with a carefully designed user interface significantly improves spatial learning in the context of wayfinding.

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